HAS-VQ: Hessian-Adaptive Sparse Vector Quantization for High-Fidelity LLM Compression

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Abstract

Post-training quantization is essential for deploying Large Language Models (LLMs) on resource-constrained devices. However, standard integer quantization (e.g., INT4) fundamentally degrades performance by imposing a uniform grid on the heavy-tailed distribution of weight parameters, particularly in smaller-scale models (e.g., $<$2B parameters). We introduce **HAS-VQ** (Hessian-Adaptive Sparse Vector Quantization), a compression framework that strictly decouples high-sensitivity outliers from the bulk weight distribution using second-order sensitivity analysis. HAS-VQ employs a **Hessian-Masked Decoupling** strategy to isolate sensitive parameters, followed by robust Vector Quantization (VQ) of the remaining dense body. Crucially, we introduce a *residual sparse feedback* mechanism that corrects quantization errors in the most sensitive dimensions, ensuring exact reconstruction of outliers. We evaluate HAS-VQ on `SmolLM2-1.7B`, demonstrating two distinct regimes of superiority: (1) **Pareto Dominance over Integer Baselines:** At 4.23 effective bits-per-parameter (BPP), we achieve a perplexity of 14.23, significantly outperforming the standard INT4 baseline (20.03 PPL at 4.71 BPP). (2) **High-Fidelity Compression:** Relative to the FP16 baseline, HAS-VQ achieves a **2.3**$\times$ reduction in model size (7.03 BPP) while maintaining statistically indistinguishable perplexity (10.12 vs. 10.04), effectively offering a lossless compression alternative for bandwidth-constrained environments.

The code is available at `https://github.com/VladimerKhasia/HASVQ`

1 Introduction

The deployment of Large Language Models (LLMs) is fundamentally constrained by the memory bandwidth wall. As model parameters scale, storing weights in half-precision (FP16) becomes prohibitive for edge devices. Post-Training Quantization (PTQ) [7, 3] has emerged as the standard solution, typically reducing bit-width via integer mapping (e.g., INT4/INT8).

However, uniform integer quantization imposes a rigid structural constraint: it assumes an equidistant discretization of the weight space. This assumption fails for the heavy-tailed, non-Gaussian distributions characteristic of Transformer weights [2, 8]. The degradation is particularly acute in smaller models (1B–3B parameters), which lack the localized redundancy to absorb quantization noise, often resulting in perplexity collapse under standard Round-To-Nearest (RTN) INT4 schemes.

Vector Quantization (VQ) offers a theoretically superior alternative by mapping blocks of weights to learned centroids in a continuous space [4], thereby aligning the codebook with the true probability density of the weights. Yet, standard VQ algorithms (e.g., K-Means) are unstable in the presence of outliers; a single high-magnitude parameter can skew centroid convergence, degrading the reconstruction of the entire weight block.

To address this, we introduce **HAS-VQ** (Hessian-Adaptive Sparse Vector Quantization). Our approach rests on the observation that while the majority of weights form a dense, quantizable body, a sparse subset of parameters possesses high Fisher Information and must be preserved with high precision to maintain the optimization trajectory [5].

Our contributions are as follows:

1. **Hessian-Masked Decoupling:** We propose a rigorous decomposition strategy that uses diagonal Hessian sensitivity to orthogonalize outliers from the dense body, ensuring that VQ centroids fit the bulk distribution without skew.

2. **Residual Sparse Feedback:** Unlike methods that simply prune or isolate weights, we implement a residual correction mechanism where the sparse component explicitly compensates for the quantization error $\mathcal{E} = \mathbf{W} - \mathcal{Q}(\mathbf{W})$ in the most sensitive dimensions.

3. **High-Fidelity Compression:** We empirically demonstrate on `SmolLM2-1.7B [1]` that HAS-VQ offers a **2.3**$\times$ reduction in storage compared to FP16 with negligible loss in fidelity (0.08 PPL difference), establishing it as a viable "near-lossless" compression format.

4. **Efficiency vs. Integer Grids:** We show strict Pareto dominance over standard INT4 baselines, reducing perplexity by 29% while requiring 11% less storage (4.23 BPP vs. 4.71 BPP).

2 Methodology

We propose **HAS-VQ**, a framework designed to minimize the Hessian-weighted quantization noise by explicitly decoupling high-sensitivity outliers from the low-frequency weight distribution.

2.1 Problem Formulation

Consider a layer weight matrix $\mathbf{W} \in \mathbb{R}^{d_{out} \times d_{in}}$. Our objective is to find a compressed representation $\widehat{\mathbf{W}}$ that minimizes the expected increase in task loss $\mathcal{L}$. Approximating the loss landscape via a second-order Taylor expansion and assuming a diagonal Fisher Information Matrix (Hessian) $\mathbf{H}$, the objective is to minimize the Hessian-weighted Mean Squared Error:

$$\min_{\widehat{\mathbf{W}}} \mathbb{E}\left[(\mathbf{W} - \widehat{\mathbf{W}})^\top \mathbf{H} (\mathbf{W} - \widehat{\mathbf{W}})\right] \approx \sum_{i,j} \mathbf{H}_{ii} (\mathbf{W}_{ij} - \widehat{\mathbf{W}}_{ij})^2 \tag{1}$$

2.2 Algorithm Design

HAS-VQ executes a four-stage compression pipeline: Sensitivity Analysis, Sparse-Dense Decoupling, Robust VQ, and Residual Integration.

2.2.1 1. Sensitivity-Based Importance Metric

To identify parameters critical to the loss landscape, we formulate an importance metric $\mathbf{I}$ that balances magnitude with curvature sensitivity. We normalize weights channel-wise by scale $\mathbf{s} \in \mathbb{R}^{d_{out}}$, yielding $\mathbf{W}_{norm}$. The importance score is defined as:

$$\mathbf{I}_{ij} = |\mathbf{W}_{norm,ij}| \cdot \sqrt{\mathbf{H}_{ii}} \tag{2}$$

This metric serves as a proxy for the change in loss $\Delta\mathcal{L}$ induced by perturbation, effectively ranking parameters by their energy contribution in the Hessian-metric space.

2.2.2 2. Hessian-Masked Decoupling

Let ρ be the sparsity ratio. We identify the set of outlier indices Ω corresponding to the top-ρ elements of $\mathbf{I}$. We decompose $\mathbf{W}_{norm}$ into a dense body $\mathbf{B}$ for quantization:

$$\mathbf{B}_{ij} = \begin{cases} 0 & \text{if } (i,j) \in \Omega \quad \text{(Masked Isolation)} \\ \mathbf{W}_{norm,ij} & \text{if } (i,j) \notin \Omega \end{cases} \tag{3}$$

By setting outliers to exactly zero in $\mathbf{B}$, we prevent the K-Means clustering from drifting toward extreme values, thereby preserving fidelity for the bulk distribution which contains the majority of the model's latent knowledge.

2.2.3 3. Robust Statistical Vector Quantization

The body $\mathbf{B}$ is blocked and clustered into codebook $\mathcal{C}$ using K-Means. To ensure rigorous convergence on limited calibration data, we employ **Stability-Bounded Sampling**, ensuring the sample size N satisfies $N \gg K$ to minimize centroid variance, alongside a **Dead Unit Revival** mechanism that relocates inactive centroids to regions of maximum reconstruction error.

2.2.4 4. Residual Sparse Feedback

A critical innovation of HAS-VQ is the handling of outliers via residual correction. Rather than simply storing the original values, we compute the *residual error* of the quantized body at the sensitive indices. Let $\mathcal{Q}(\mathbf{B})$ be the quantized reconstruction of the body. The sparse correction $\mathcal{S}$ is defined as:

$$\mathcal{S}_{ij} = \mathbf{W}_{norm,ij} - \mathcal{Q}(\mathbf{B})_{ij}, \quad \forall (i,j) \in \Omega \tag{4}$$

This ensures that the final reconstruction at sensitive indices is mathematically exact (up to FP16 precision):

$$\widehat{\mathbf{W}}_{ij} = \mathcal{Q}(\mathbf{B})_{ij} + \mathcal{S}_{ij} = \mathbf{W}_{norm,ij} \tag{5}$$

This feedback loop effectively eliminates quantization noise in the dimensions with the highest Fisher Information.

Algorithm 1 HAS-VQ Compression Pipeline

Input: $\mathbf{W}$, Hessian $\mathbf{H}$, sparsity ρ, blocks b, centroids K
$\mathbf{s} \leftarrow \max(|\mathbf{W}|, \dim = 1); \quad \mathbf{W}_{norm} \leftarrow \mathbf{W}/\mathbf{s}$
$\mathbf{I} \leftarrow |\mathbf{W}_{norm}| \odot \sqrt{\mathbf{H}}$ ▷ Compute Sensitivity
$\Omega \leftarrow \mathrm{TopK}(\mathbf{I}, \rho)$ ▷ Identify sensitive indices
$\mathbf{B} \leftarrow \mathbf{W}_{norm}; \quad \mathbf{B}[\Omega] \leftarrow 0$ ▷ Masked Decoupling
$\mathcal{C}, \mathcal{IDX} \leftarrow \mathrm{RobustKMeans}(\mathbf{B}, K, b)$
$\mathbf{R} \leftarrow \mathcal{C}[\mathcal{IDX}]$ ▷ Reconstruction
$\mathcal{S} \leftarrow (\mathbf{W}_{norm} - \mathbf{R})[\Omega]$ ▷ Compute exact residual
return $\mathbf{s}, \mathcal{C}, \mathcal{IDX}, (\Omega, \mathcal{S})$

3 Experiments

We evaluate HAS-VQ on `SmolLM2-1.7B-Instruct`. We compare effective Bits-Per-Parameter (BPP)—calculated as the aggregate storage of codebooks, indices, scales, and sparse residuals divided by parameter count—against Perplexity (PPL) on `wikitext-2 [6]`.

3.1 Experimental Results

Table 1 details the performance. The compression ratio (CR) is calculated relative to the 16-bit half-precision baseline ($CR = 16.0/\text{BPP}$).

3.2 Analysis

Pareto Dominance over Integer Constraints The **HAS-VQ (Mid)** configuration demonstrates strict Pareto dominance over the standard INT4 baseline.

1. **Accuracy:** It reduces Perplexity from 20.03 (INT4) to 14.23, recovering significant linguistic capability.
2. **Efficiency:** It achieves this with a smaller memory footprint (4.23 bits) than the INT4 baseline (4.71 bits).

This result confirms that the VQ manifold, when augmented with Hessian-guided residuals, provides a superior rate-distortion curve compared to the rigid integer grid inherent to standard quantization baselines.

Table 1: Results on SmolLM2-1.7B. **BPP** represents total effective bits per parameter. **Ratio** is relative to FP16. HAS-VQ exhibits two operating points: Pareto dominance over INT4 and High-Fidelity compression comparable to FP16.

Method	**Configuration**	**PPL** $\downarrow$	**BPP** $\downarrow$	**Ratio** $\uparrow$	**Regime**
FP16	Oracle	10.04	16.00	1.0x	Lossless
INT4	RTN Baseline	20.03	4.71	3.4x	Lossy (Degraded)
HAS-VQ	**Mid (Balanced)**	**14.23**	**4.23**	**3.8x**	**Efficient Dominance**
HAS-VQ	**High (Fidelity)**	**10.12**	**7.03**	**2.3x**	**Near-Lossless**

High-Fidelity Compression (The 7 BPP Regime) Crucially, the **HAS-VQ (High)** experiment demonstrates that our method can serve as a near-lossless compression format. By increasing the bit-budget to 7.03 BPP, HAS-VQ achieves a perplexity of 10.12, which is statistically indistinguishable from the Oracle FP16 perplexity of 10.04. This represents a **2.3**$\times$ reduction in model size without the performance penalties typically associated with quantization. This result is significant for bandwidth-limited deployment scenarios where model accuracy cannot be compromised, yet storage capacity is finite.

4 Discussion

4.1 Interpretation of Results

Our experimental results highlight the limitations of integer-based quantization for small-scale LLMs. The Gaussian-like distribution of the weight "body" is inefficiently mapped by uniform integer grids. HAS-VQ addresses this by allocating bits via density-matching (VQ) and explicitly handling high-curvature weights via the sparse residual stream. The ability to recover FP16-level performance at 7.03 BPP suggests that the intrinsic dimension of the weight distribution is significantly lower than 16 bits, provided the quantization noise is spectrally shaped to avoid high-Hessian directions.

4.2 Architectural Generality

The mathematical formulation of HAS-VQ relies solely on linear algebra operations—specifically the Singular Value Decomposition (implicit in Hessian approximation) and K-Means clustering—rather than Transformer-specific heuristics. Consequently, this method is theoretically applicable to any architecture dominated by linear layers, including Mixture-of-Experts (MoE) and Convolutional Neural Networks (CNNs).

4.3 Implications for Edge Deployment

The storage savings offered by HAS-VQ enable the deployment of 1.7B-scale models on devices with highly constrained memory budgets. The *High Fidelity* configuration (7.03 BPP) is particularly relevant for scenarios requiring "download-and-run" efficiency, reducing the data transfer requirement by over 50% compared to FP16 weights while ensuring the end-user experiences the full capability of the uncompressed model.

5 Conclusion

We presented HAS-VQ, a method utilizing Hessian-Masked Decoupling to achieve rigorous outlier separation. Our results on SmolLM2-1.7B are twofold: (1) We establish a new Pareto frontier relative to integer baselines, outperforming INT4 in both size (-11%) and accuracy (+29% PPL improvement); and (2) We demonstrate a high-fidelity operating point at 7.03 BPP that matches FP16 performance with 2.3$\times$ compression. These findings suggest that HAS-VQ is a robust candidate for both aggressive compression and high-fidelity storage reduction in edge AI applications.

References

[1] Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, Colin Raffel, Leandro von Werra, and Thomas Wolf. Smollm2: When smol goes big – data-centric training of a small language model, 2025.

[2] Tim Dettmers, Mike J. G. Lewis, Younes Belkada, and Luke Zettlemoyer. Gpt3.int8(): 8-bit matrix multiplication for transformers at scale. In *Neural Information Processing Systems*, 2022.

[3] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. Gptq: Accurate post-training quantization for generative pre-trained transformers. *ArXiv*, abs/2210.17323, 2022.

[4] Robert M. Gray. Vector quantization. *IEEE ASSP Magazine*, 1:4–29, 1984.

[5] Yann LeCun, John S. Denker, and Sara A. Solla. Optimal brain damage. In *Neural Information Processing Systems*, 1989.

[6] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models, 2016.

[7] Markus Nagel, Rana Ali Amjad, Mart Van Baalen, Christos Louizos, and Tijmen Blankevoort. Up or down? Adaptive rounding for post-training quantization. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7197–7206. PMLR, 13–18 Jul 2020.

[8] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Neural Information Processing Systems*, 2017.